

SWARM-SLR - Streamlined Workflow Automation for Machine-Actionable Systematic Literature Reviews

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Abstract. Authoring survey or review articles still requires significant tedious manual effort, despite many advancements in research knowledge management having the potential to improve efficiency, reproducibility, and reuse. However, these advancements bring forth an increasing number of approaches, tools, and systems, which often cover only specific stages and lack a comprehensive workflow utilizing their task-specific strengths. We propose the Streamlined Workflow Automation for Machine-actionable Systematic Literature Reviews (SWARM-SLR) to crowdsource the improvement of SLR efficiency while maintaining scientific integrity in a state-of-the-art knowledge discovery and distribution process. The workflow aims to domain-independently support researchers in collaboratively and sustainably managing the rising scholarly knowledge corpus. By synthesizing guidelines from the literature, we have composed a set of 65 requirements, spanning from planning to reporting a review. Existing tools were assessed against these requirements and synthesized into the SWARM-SLR workflow prototype, a ready-for-operation software support tool. The SWARM-SLR was evaluated via two online surveys, which largely confirmed the validity of the 65 requirements and situated 11 tools to the different life-cycle stages. The SWARM-SLR workflow was similarly evaluated and found to be supporting almost the entire span of an SLR, excelling specifically in search and retrieval, information extraction, knowledge synthesis, and distribution. Our SWARM-SLR requirements and workflow support tool streamlines the SLR support for researchers, allowing sustainable collaboration by linking individual efficiency improvements to crowdsourced knowledge management. If these efforts are continued, we expect the increasing number of tools to be manageable and usable inside fully structured, (semi-)automated literature review workflows.

Keywords: Systematic Literature Review · Workflow Automation · Requirement · Software Tool · Crowdsourcing

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1 Introduction

A key pillar of the scientific method is the synthesis of various research contributions in survey or review articles. The number of reviews claiming to be "comprehensive", however, declined over the past two decades, with the number of relevant publications rapidly increasing [28]. Larsen et al. [28] show that the current practice of authoring surveys does not scale to this increase in workload, is error-prone, and lacks reproducibility and reuse. Even in fields like medicine, where systematic protocols were established for meta-analysis, authors like Ioanidis [21] conclude that there is mass production of redundant, misleading, and conflicting systematic reviews and meta-analyses.

Systematic Review Automation (SRA) aims to utilize "computational assistance and automation to dramatically improve the process" while being "faster, with fewer resources" [50]. With software solutions becoming more accessible while data machine-readability increases, the future of scholarly knowledge discovery and distribution can be greatly improved [22]. Evidently, this productivity acceleration can reduce the required time from months to only two weeks, while apparently maintaining or even increasing scientific integrity [17]. Artificial Intelligence (AI) further promises to automate repetitive tasks and support others in so-called AI-based literature reviews (AILRs) [51].

These advancements face reservations and doubts, as they further highlight the requirement of "human interpretation and insightful syntheses, as well as novel explanation and theory building." [51, p. 13]. With Large Language Models (LLM) promising a new interface to digital libraries, information retrieval, and knowledge synthesis become more accessible to a broader user group. These user groups may struggle to maintain their domain integrity when faced with unreliable results caused by the inherent hallucination of LLMs and misunderstood processes [7]. The new tools with hallucinating capabilities potentially challenge the current perception of tool usage, where the user is not to be perceived as the author, but a commissioner [2]. The Texas University Library specifically lists ChatGPT as an "AI-Based Literature Review Tool" [56]. These promises follow certain expectations: Any Literature Review Tool should only review existing literature, identify relevant evidence, and help accurately represent the state of the art. Since at least some of these new tools don't satisfy these fundamental requirements, it follows that not every Literature Review Tool satisfies every requirement. This warrants a thorough definition of these requirements and a transparent tool capability assessment. In this work, we specify an initial set of requirement definitions alongside several tool assessments to provide education on capabilities and limitations.

For a complex, multistaged process such as a Systematic Literature Review, a workflow spanning multiple tasks and tools can become a daunting task. Some accessible literature review workflows chaining these new tools already exist [43], lowering this barrier to entry. These workflows have the potential to educate on proper tool usage, leading to standardization and improved efficiency and quality, When facing an exponentially growing scholarly knowledge corpus [13], an SLR workflow should enable researchers across domains to collaboratively structure knowledge Findable, Accessible, Interoperable, and Reusable (FAIR) [55].

With this work, we provide insights into how software tools can be used to systematically reduce inconsistency by streamlining an accessible literature review process toward crowd- or expert-sourced knowledge management.

Our contributions include: (1) We synthesize SLR guidelines into 65 requirements for the **S**treamlined **W**orkflow **A**utomation fo**r** **M**achine-actionable **S**ystematic **L**iterature **R**eviews (*SWARM-SLR requirements*), allowing distributed collaboration and development of cohesive and comprehensive SRA. (2) We propose the ready-to-use *SWARM-SLR workflow* spanning interoperable SRA from planning to reporting a review, demonstrating the usage of software tools situated according to their strengths and weaknesses (*SWARM-SLR tools*).

To achieve this, we define the scope of SWARM-SLR in Sect. 2. With that groundwork laid out, we refine the requirement and tool sets in Sect. 3 to synthesize them into a workflow. The requirements, tools, and workflow are evaluated in Sect. 4 and discussed in Sect. 5, allowing us to draw a promising conclusion about the future of SWARM-SLR in Sect. 6.

2 Related Work

We consider related work regarding Systematic Literature Reviews (SLR) and Software Tools in their significance for Systematic Review Automation (SRA).

Systematic Literature Reviews (SLR) aim to identify, analyze, synthesize, and report literature [48]. There are many categorizations of different literature review types [6,39]. According to Adams et al. [6], review types may be categorized into "Instrumental Review", "Evaluative Review" and "Exploratory Review", of which SWARM-SLR aims to improve primarily the latter. The individual approach varies depending on domain, context, and goal, which is why our approach focuses on answering specific research questions, one of five purposes of a literature survey according to Torraco [48]. The author also states that "the body of literature on literature reviews has grown from a few publications to a large body of literature", covering often interchangeably used terms like "literature review", "integrative literature review", "research review" and "review of the literature", while the term "review" is often substituted for "survey". For the context of this paper, SWARM-SLR is domain-independently targeted at researchers conducting either a literature survey or review, especially when conducting an exploratory review answering specific research questions.

Software Tools of various degrees of reliability, particularly with the general advances of AI, have been introduced into scientific processes [43], surfacing novel challenges in AI literacy education to maintain integrity [30]. Especially since tools like Large Language Models (LLM) "display wider scope and autonomy", [36, p. 5] there is a wide array of misconceptions about what these tools can do and where the user has to supervise the results themselves. While there are risks inherent to handing off responsibility to these tools, other dedicated survey support software has been developed, increasing the efficiency while maintaining integrity. Open Knowledge Maps (OKMaps) [1,26] specifically addresses accessible user engagement in search processes and focuses on reliable discovery. Similar

web search assistant tools like Connected Papers [5] and long-established services like Google Scholar or Semantic Scholar also address these problems [29,33]. Many of these tools integrate with one another. ArXiv [4] for example has various references and embeds Connected Papers on its website, which again builds on Semantic Scholar. This redundancy allows users to easily verify information retrieved at one location with other sources. Meta services like the Open Research Knowledge Graph (ORKG) [44] can make use of this traceable knowledge journey as a core tool in the literature review process, especially in the final steps. The Best Practices in Data Science and Artificial Intelligence by Borisova et al. [12] specifically recommend using the ORKG, especially in combination with semantic annotation through SciKGTeX [10]. This specialization of dedicated tools, capable of reliably solving certain tasks, stands in contrast to the general capabilities of recent conversational agents without certainty in any area. This leads to an incoherent spread of tools across the workflow. To situate each tool where it excels, we need to understand what work needs to be done in the first place.

Systematic Review Automation (SRA) is an ongoing topic in scientific discourse [25,32,49–52]. The tools developed here span certain stages of the SLR, sometimes fully automating certain tasks, while others are semi-automated with a tool-assisted human-in-the-loop approach. Particularly, AI-based Literature Reviews (AILR) are highlighted for their potential by Wagner et al. [51]. Wagner et al., Kohl et al. [25], as well as Pulsiri and Vatananan-Thesenvitz [38] all formulate different frameworks for assessing different capabilities of SLR tools, naming different tasks and evaluating the fitness of tools differently. Pulsiri and Vatananan-Thesenvitz for example created an overview on automating the SLR. They adapt the structure illustrated in Table 1 to analyze how and where the state-of-the-art SLR can be automated.

Table 1. Mapping papers in relation to automation methods and SLR according to Pulsiri and Vatananan-Thesenvitz [38].

Stage	Task	Semi-automated	Fully automated
I	*Planning a review*	[34,49,50]	
	Defining a scope	[34,49,50]	
II	*Search*	[34,41,46]	[18,24,49]
	Select	[34,37,41,46,47,53,54]	[18,49,50]
	Evaluate	[34,46]	[18,49,50]
III	*Analysis*	[34,35,46,49,50]	
	Synthesis	[34,46,49,50]	
IV	*Reporting*	[34,46,49,50]	

This table has the benefit of being tailored to tools supporting the review process, similar to online registers such as "Systematic Review Toolbox" with over 100 tools, 78 of which apply to the systematic literature review [3]. These tools, however, are mostly not standalone applications that can be readily applied to any scientist's workflow, lacking interfaces to other tools along the chain or even a cohesive workflow spanning the entire SLR. Bosman and Kramer [14,15] compiled an extensive work by collecting over 700 of these tools. While not mapped directly to the review tasks displayed in Table 1, they are categorized into 30 phases of scholarly communication, many of which can be situated inside the literature survey process. A selection of these tools has also been classified according to their fitness for 17 specific tasks, along with a dedicated survey with 20,000 respondents solidifying this mapping [16]. Similarly, research-paper recommender systems were reviewed by Beel et al. [8] and Kreutz and Schenkel [27]. In total, they provide an overview of over 150 paper recommender systems, as well as two examples of how SLR might be conducted. All of these tools provide a strong set to choose from when designing a framework for the SWARM-SLR while highlighting the need to streamline a cohesive workflow to navigate them efficiently.

We have now established the elusive definition of the literature review process, as well as the sheer abundance of tools addressing their interpretation and scope. What they lack is a consistent, fundamental framework of what tasks and steps an SLR requires. Since currently no tool can achieve SRA alone, their strengths and weaknesses need to be comparable over the entire scope. Given this plethora of options, it is very likely that task-excelling tools combined into a modular workflow would make SRA more achievable than any currently available tool alone. Our work builds towards this modular framework, connecting available tools with automated interfaces where possible, and human-in-the-loop supervision covering segments where full automation of sufficient quality is not yet achievable.

3 Approach

To properly describe the approach described in Fig. 1, we need to define *SWARM-SLR requirements*, *SWARM-SLR tools* and *SWARM-SLR workflow*.

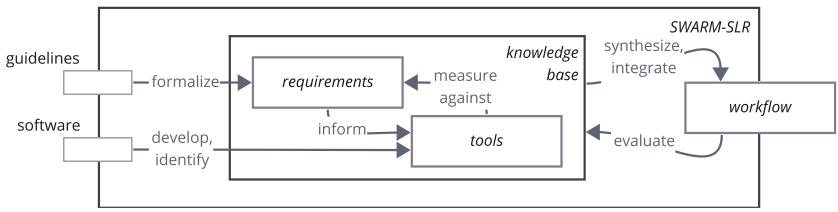


Fig. 1. Formalizing SWARM-SLR requirements informs the identification and development of the tool catalog. Synthesizing these into a tool-assisted workflow makes this knowledge actionable and enables evaluation.

Requirements. To realize streamlined workflow automation, it is advisable to formalize requirements. Requirements enable the comparison of tools and workflows by allowing us to assess them against a common set of objectives. These objectives may be structured into groups and should formally represent the stages and tasks of a systematic literature review. In this context, *SWARM-SLR requirements* describe a set of requirements formalized from SLR literature, specifically with the intent to assess the applicability of *SWARM-SLR tools*.

Tools. A tool enables solving a given task at a higher efficiency or effectiveness. In the context of scientific workflow automation, specifically for literature reviews, tools are usually software tools, but categorically they can be of any nature. While chaining software tools into a toolchain is easier due to interoperability, building a tool-assisted workflow implementing non-software physical tools is also possible. In this context, *SWARM-SLR tools* describe a set of tools measured against the *SWARM-SLR requirements*, with the potential to be used as a tool catalog for the *SWARM-SLR workflow*.

Workflow. A workflow structures required operations alongside tools and recommendations to achieve a higher efficiency or effectiveness. In this context, the *SWARM-SLR workflow* describes guidance alongside the *SWARM-SLR requirements* while implementing and interconnecting *SWARM-SLR tools*.

With an efficient SWARM-SLR workflow, future SLR have the potential to scale against the increasing amount of publications each year [13] depicted in Fig. 2. By integrating the semantification of existing literature in a collaborative environment, each SWARM-SLR review potentially contributes to one another, reducing the workload of future research regarding pre-existing literature.

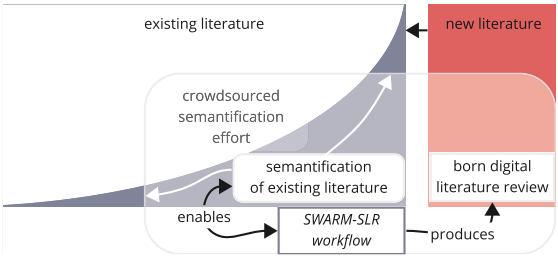


Fig. 2. The SWARM-SLR workflow enables and is enabled by the semantification of literature. It also produces a born-digital literature review, catalyzing future knowledge synthesis.

The approach was iteratively developed following the design science methodology using a demo survey case of aerospace engineering knowledge representations. This enabled us to refine certain requirements and test a first selection of tools. Simultaneously, these explorations allowed us to integrate additional requirements and tools, either arising by the process itself or suggested from the surveyed literature, for example, the Natural Language Processing (NLP)

pipeline by Berquand [9]. The respective development process and result are presented in the subsections Requirements, Tools and Workflow.

3.1 Requirements

Using the baseline of Pulsiri and Vatananan-Thesenvitz, we used guidance from Wagner, Lukyanenko, Templier and Paré [45,51], particularly for the first three stages (see Table 1). We detailed each task into individual steps, results, and requirements with basic literature like the Cochrane Handbook [19] and especially Machi and McEvoy [31], a shared source of all the aforementioned authors.

This version and its resulting requirements were iteratively discussed among authors and fellow scholars, especially from the fields of NLP and digital libraries. Once the coverage was optimized to the best of the authors' knowledge, these requirements were properly formalized according to the guideline by Rupp [42]. In the end, we managed to detail the eight tasks from Table 1 into 65 specific requirements, as shown in Tables 2, 3 and 4.

Table 2. SWARM-SLR requirements of stage I.

Task	Step	Result	ID	**Requirement:** The tool should be able to help the user to efficiently ... for the <result>.
1. Planning a review	*1. Formulate research interest*	explicit informal research interest	1	formalize abstract interest
			2	utilize central keywords
2. Defining a scope	*1. Check for related research questions*	narrowed informal research interest	3	know what previous scholarly works already covered
			4	know what previous scholarly works were missing
	2. Refine scientific interest	specific research question	5	use specific vocabulary
			6	state a clear scope
			7	state a clear perspective
	3. Formulate Search Query	preliminary weighted keywords and queries	8	use relevant keywords
			9	weight individual keywords
	4. Refine with related literature	weighted keywords and refined queries	10	extend the vocabulary
			11	identify polysemes and synonyms
	5. Reevaluate with domain experts	validated weighted keywords and queries	12	validate research questions
			13	validate keywords
			14	validate weights
			15	validate search query

3.2 Tools

Given the requirement set, the next task was to identify the most suitable tools. Hundreds of tools would be applicable here, as illustrated in Sect. 2 Systematic Review Automation (SRA). Many tools redundantly cover the same

Table 3. SWARM-SLR requirements of stage II

Task	Step	Result	ID	**Requirement:** The tool should be able to help the user to efficiently ... for the <result>.
3. Search	*1. Find resources*	preliminary document set	16	use reliable sources
			17	find relevant documents
			18	find similar documents
	2. Remove duplicates and find missing documents	curated document set	19	identify duplicate documents
			20	find unindexed documents
			21	identify document set gaps
4. Select	*1. Extract structured data from documents*	metadata	22	extract publication date
			23	extract author(s)
			24	extract publication venue
			25	extract specified keywords
		content	26	extract document text
			27	differentiate chapters
		bag of words	28	remove special character
			29	identify multi-words
			30	expand acronyms
			31	lemmatize words
			32	normalize words
			33	remove stop words
		contribution statements	34	identify statements
	2. Calculate relational measurements within the document set	document representation	35	calculate "term frequency" (tf) and "term frequency - inverse document frequency" (tf-idf)
			36	calculate word embedding and document embedding
			37	represent document machine-readable
		similarity of documents within the document set	38	consider synonyms
			39	consider polysemes
			40	calculate document similarity
	3. Identify documents relevant for the research questions	relevance of documents for research question	41	represent research question machine-readable
			42	calculate document relevancy for research question
5. Eval -uate	*1. Remove out-of-scope document subset*	in-scope document set	43	define in-/exclusion criteria
			44	exclude documents
	2. Evaluate relevancy measurement	selection approval	45	evaluate similarity of documents
			46	evaluate relevance of documents
	3. Classify document subsets	literature subsets	47	calculate variables
			48	divide document set into subsets

task, which can be beneficial to quality and completeness, but simultaneously detrimental to runtime and complexity. Many tools focus on wide breadth, while others go into much more depth for only specific tasks. We began evaluating these tools using our requirements set alongside developing it, further improving both sets.

Table 4. SWARM-SLR requirements of stage III and IV

Task	Step	Result	ID	**Requirement:** The tool should be able to help the user to efficiently ... for the <result>.
6. Analysis	*1. Annotate resources*	contribution representation	49	annotate the document
			50	represent the annotation machine-readable
			51	annotate collaboratively
7. Synthesis	*1. Compare contributions*	comparison	52	define comparison properties
			53	insert contribution values
	2. Craft arguments based on findings	document claims	54	identify evidence (data)
			55	establish warrant
	3. Critique the literature	thesis claims	56	identify evidence (claims)
			57	identify patterns
			58	detect fallacies
			59	identify conflicts
			60	resolve conflicts
8. Reporting	*1. Write the literature review*	review article	61	utilize comparisons
			62	support claims
			63	establish outline
			64	draft iteration(s)
			65	audit iteration(s)

Finally, we selected a preliminary subset to begin working on an exemplary workflow. We selected according to the following criteria: Addressing SRA steps formalized by the requirements (e.g. Colandr or Cochrane RevMan); Known tool compatibilities, i.e. the ability to link a tool output to a consecutive input (e.g. Connected Papers to Zotero); The user community, i.e. the degree of adoption already available (e.g. Google Scholar or ChatGPT); Previous experience with the tools (e.g. Python and its NLP libraries).

Likely, all of these tools have equivalents, competitors, and complementing software outside our current subset. By focussing on the modularity provided by the requirement framework, any such tools can be slotted accordingly, if sufficient adapters are existing or developed. Our subset used for the exemplary workflow development is depicted in Fig. 3.

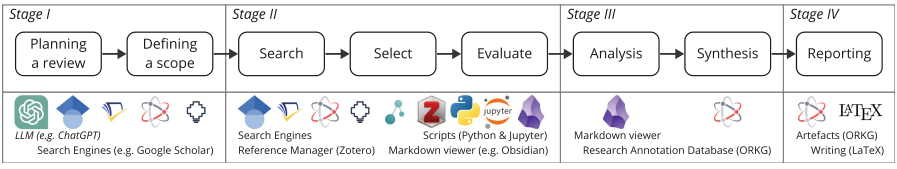


Fig. 3. SWARM-SLR workflow guided by four online Jupyter Notebooks.

While *CADIMA* was of exceptional interest as it structures the SLR, its rigid structure set it on the back of the list, providing a guideline similar to the requirements, but not an implementation priority by itself. Using *Large Language Models (LLM) like ChatGPT or Llama*, however, was exceptionally incentivized due to their rising popularity, simultaneously leveraging new potential as well as providing SWARM-SLR users evidence and education regarding LLM limitations. *Google Scholar* and *Semantic Scholar* were included due to their exceptional search capability, while *Open Knowledge Maps (OKMaps)* and *ConnectedPapers* complimented them without significantly increasing the overhead. *Zotero* was chosen for reference management due to being free and widely used in the community, providing an interface from search results to a local library. *Python*, implementing *NLP libraries like NLTK, spaCy, FLAIR* and *PDF miners like pdfminer, pypdf2*, was used to analyze the document set. While using *Open Research Knowledge Graph (ORKG)* alongside *SciKGTeX* was recommended by the Best Practices in Data Science and Artificial Intelligence [12], the ORKG seemed particularly suitable for storing both intermediate and final products of the workflow. For publication and formatting, SciKGTeX leads to the selection of *LaTeX*, amongst other interoperability benefits. Despite not being listed specifically as SWARM-SLR tools, supplementary tools like a coding IDE (*Visual Studio Code*), a markdown viewer (*Obsidian*), or PDF viewer (*Acrobat Reader*) will be used by most users.

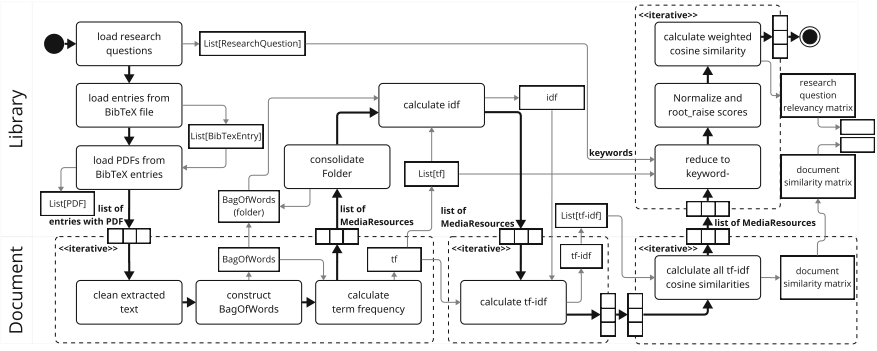


Fig. 4. UML activity diagram for the fully automated *Task 4: Select*

3.3 Workflow

The workflow is composed of four Notebooks connecting the tools using a Python Package. Printing the GitHub representation of all notebooks, including examples, code snippets, LLM prompts, and replies, as well as illustrations, would comprise over 25 pages. We refer to the repository for details.[1]

Package. To automate certain tasks, we built and released the open source python package bnw_tools [11]. The package implements tool interfaces, analyzing Zotero entries to produce insights in Obsidian readable files, using PDF miners and NLP. As Fig. 4 illustrates, it can automate certain tasks like *Task 4: Select* when it is installed and executed in the local library.

Notebooks. Four Jupyter Notebooks guide through the workflow as depicted in Fig. 3, each containing the markdown guideline, tool support, and code for the respective stage in the SWARM-SLR workflow.[1] They are optimized for GitHub and can be used as an online reference with no prior technical expertise required. Using checklists, visualizations, and step-by-step instructions, they assist the user in navigating through each requirement and completing each individual task of an SLR. Figure 5 illustrates the visualization during data discovery while showing the machine-readable representation of a research question with weighed keywords, queries, and dynamically generated lists of relevant documents.

Similarly, the literature further processed by Task 6 and 7.1 eventually create ORKG representations, which both directly contribute to the current SWARM-SLR in the form of comparisons as shown in Fig. 6, as well as to future SLR due to the persistent FAIR representation. Taking the recommendation systems SLR by Kreutz and Schenkel [27] as an example, a template[2] allows the transition from a table bound to two pages in a PDF publication to an interoperable knowledge graph. By implementing a subset of the Shapes Constraint Language (SHACL) [20], representations like these enable interoperable reuse, comparison[3], and future expansion. This results in dynamic reviews founded on connected knowledge and literature management, supporting asynchronous collaboration beyond the scope of single, disjoint articles [23].

4 Evaluation

This section presents the survey Design and Results to evaluate SWARM-SLR.

4.1 Design

Based on the validation guidance by Robson and McCartan [40], we designed a tandem-survey to validate SWARM-SLR: One survey[4] validates which require-

[1] https://github.com/borgnetzwerk/tools/tree/main/scripts/SWARM-SLR.
[2] Recommendation system template: https://orkg.org/template/R673347/.
[3] Recommendation systems comparison: https://orkg.org/comparison/R674188/.
[4] Requirement review survey: https://survey.uni-hannover.de/index.php/555283.

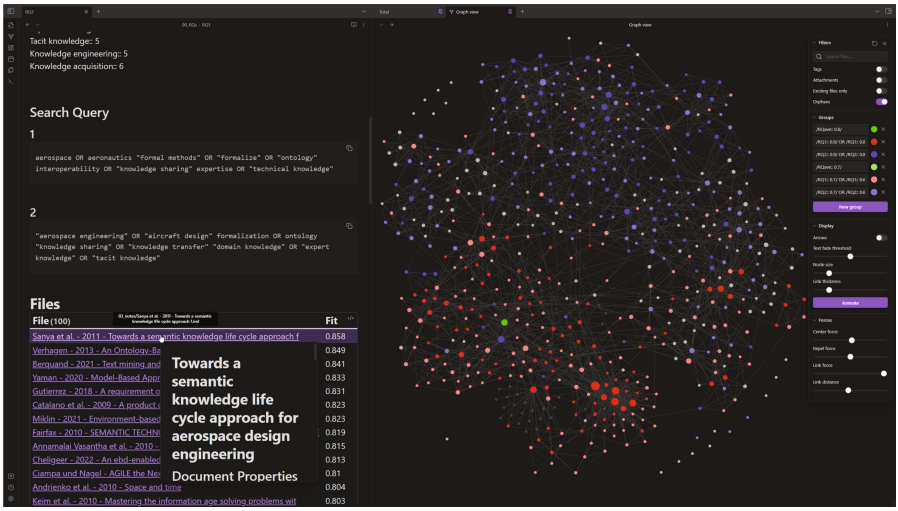


Fig. 5. Obsidian overview of the analyzed library. The graph represents documents as nodes connected to similar documents and colored based on their relevance to a research question (RQ) (here: RQ1 red, RQ2 blue, average green). The RQ1 file on the left lists its dynamically ranked most relevant documents. (Color figure online)

ments correctly represent SLR tool requirements, the other[5] used these requirements to gauge the fitness of a given tool to a given task.

The modular structure provided through Rupp [42] made the formulation of survey questions simple: from "*Requirement: The tool should be able to help the user to efficiently <requirement> for the <result>.*", we derived the statement template "*To complete the step '<step>', a tool should be able to help the user to efficiently <requirement> for the <result>.*". "*a tool*" was replaced with "*the tool*" for the tool survey, which begins with a selection from 29 SRA tools or a free-form text field to specify which tool would be assessed.

The respondents indicated their agreement with each statement on a 9-point Likert scale from "strongly agree" (A1) to "strongly disagree" (A9), with the option to not answer (A10). All steps of the same task were listed on the same page, headlined by the task. These eight pages were framed by an opening page providing context, a page for demographic and literature survey expertise data, a free-form feedback question, and a closing page pointing towards the respective other survey. The feedback question specifically requested "*Would you add any missing tasks, steps or requirements? If so, where?*", to aid in expanding this preliminary requirement set. The entire survey structure, including every question and answer option, is available for reference[6].

[5] Tool assessment survey: https://survey.uni-hannover.de/index.php/628237.
[6] https://github.com/borgnetzwerk/tools/blob/main/scripts/SWARM-SLR/data/.

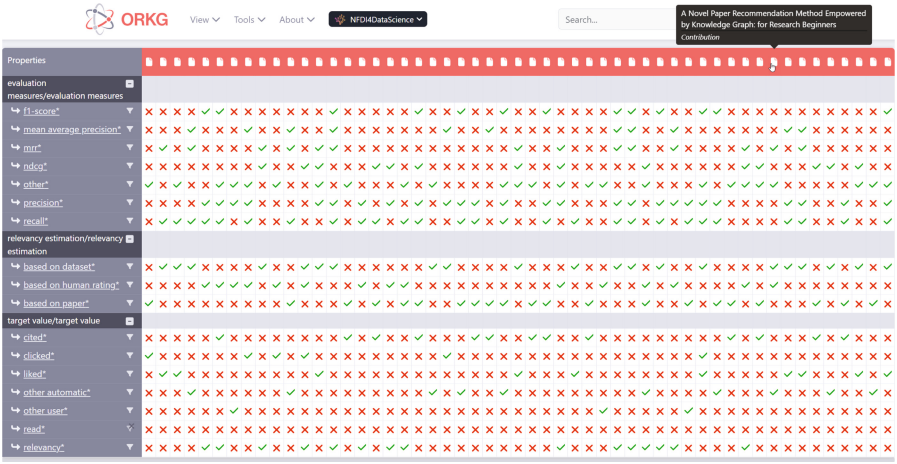


Fig. 6. Scientific paper recommendation systems literature review by Kreutz and Schenkel [27], represented in the ORKG, visualized as an ORKG comparison (see footnote 3).

The links to the surveys were distributed in mid-2023 and early 2024, iterating on the design of the workflow in between. In 2023, the surveys were shared directly with over 200 fellow researchers, primarily from the computer science and engineering domains. Additionally, the surveys were distributed over outreach channels such as LinkedIn and SurveyCircle. In 2024, the link to the tool survey was shared again with the 200 researchers, now with the added information where the SWARM-SLR workflow can be found and how it can be evaluated by selecting "other", then "SWARM-SLR" on the tool selection page. For a long-term sustainable evaluation of these requirements and tools like Bosman and Kramer displayed, we have built an evaluation workflow around it, allowing for the continuous instream of differentiated evaluations, including new tool suggestions, added requirements and comments (see footnote 6). All data processing algorithms, details and results are accessible there.

4.2 Results

The required expertise to provide a valuable contribution to the topic in question proved to be a significant obstacle. When faced with the depth of requirements, many respondents highlighted that the different tasks require very different domain knowledge. Stage I appears to lean towards science management, stage II and III towards information retrieval, and stage IV towards scholarly communication. In total, the requirement assessment was completed by 11 respondents, all of which were valid responses. The tool assessment was completed by 37 respondents, of which 36 were valid. Overall, 12 tools have been evaluated, 6 of them by multiple respondents. The SWARM-SLR Workflow has been evaluated

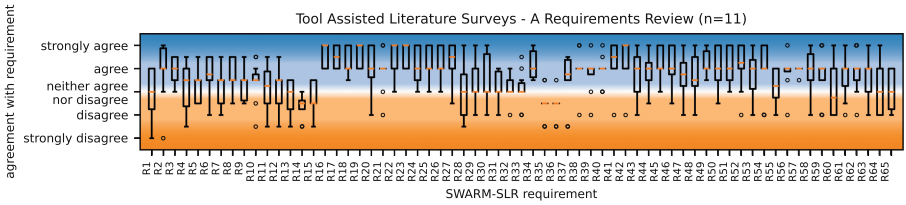


Fig. 7. Box plot of the agreement with the 65 requirements, as reported in the Requirements Review survey.

by 11 respondents, for 40 min on average ($max = 60, min = 13, \sigma = 17.63$). Two of these were from the engineering domain and nine from computer science.

Requirements. Figure 7 displays the plotted agreement over all 65 requirements. The general agreement is rather positive, with selected exceptions like the re-evaluation with domain-experts (R12-R15, Table 2) or the tf-idf and word-/document-embeddings (R35-R36, Table 3).

When looking at expertise reported in the demographics (see footnote 6), non-experts were extremely skeptical of tools supporting a re-evaluation with domain experts (R12-R15), while reported experts judged neutrally reserved and with balanced agreement. A particularly interesting finding is that calculating tf-idf and word-/document-embeddings (R35-R36) was deemed overall not required. Being probably the most technical requirements in the entire set, it is possible that these requirements are too abstract to be interpretable by general survey participants. This assumption is particularly reinforced when observing that requirement R40 "calculate document similarity" is agreed on, while both *tf-idf* and word embeddings are components of state-of-the-art document similarity analysis.

Overall, this evaluation identifies reserved skepticism towards a tool supporting *Step 2.5 Reevaluate with domain experts* (R12-R15). The vast majority of requirements, particularly in stages II through IV (R16-R65), are sustained and appear to properly reflect what tool support is required for SWARM-SLR. Over both the requirement and tool survey combined, no additional task, step or requirement was reported missing from the set.

Tools. To interpret Fig. 8, we can set the highest values reported in the tool review as an upper boundary currently perceived possible. This value may be biased due to subjective interpretation of the requirements, the tools' capability, or similar inflation. The more respondents report for a tool, the clearer the picture, and the easier it is to identify outliers. Google Scholar, for example, has been evaluated 10 times and clearly reflects its strength in search-related tasks. LLMs similarly have been assessed 3 times, showing clear strengths in a wide

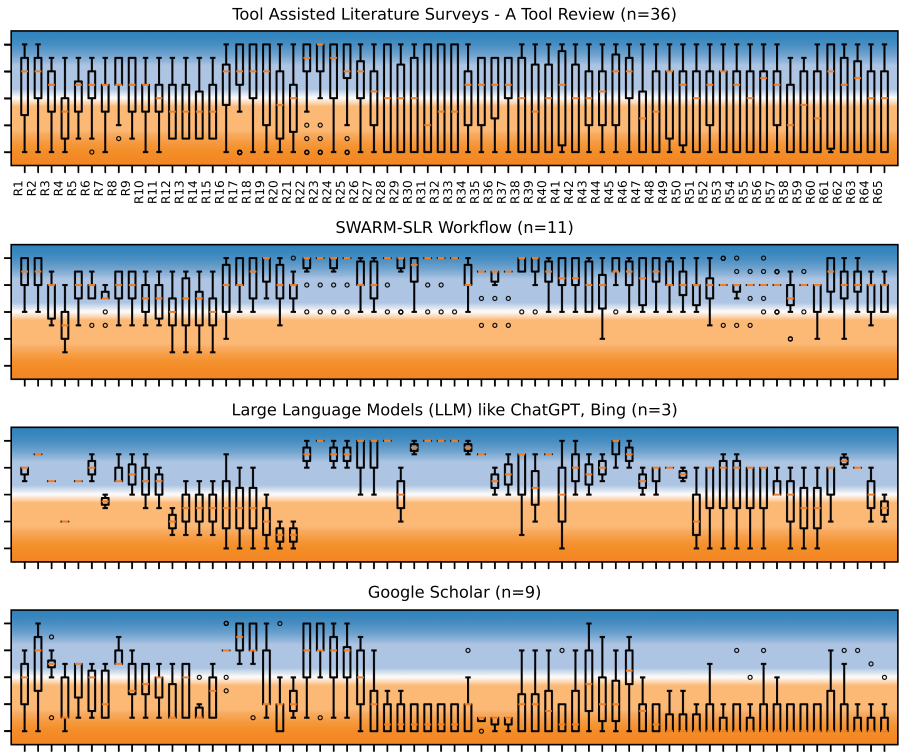


Fig. 8. Box plots of the agreement with the tools (All, SWARM-SLR Workflow, LLM, Google Scholar) supporting the 65 SWARM-SLR requirements. Identical axis labels to Fig. 7, reduced for better viewability.

range of tasks, while simultaneously showing strong limitations and skepticism in many of them. Finally, the SWARM-SLR workflow has been more thoroughly evaluated than all other tools combined (total: 442 min, median: 39 min vs. total: 386 min, median: 10 min). It was able to streamline the workflow while apparently addressing most of the requirements, with some reservations at *Step 2.5 Reevaluate with domain experts* (R12-R15) as well as in *Step 7.3 Critique the literature* (R58).

Outside the requirements survey, the SWARM-SLR workflow tool evaluation was the only one that included comments. Noteworthy is a positive sentiment ("*Can't wait to test SWARM on my next literature review. Seems like a thoroughly thought-through methodology. Cheers to the development!*", "*The approach is structured in a nice way and has a comprehensive list of tasks. Which is good to have a detailed step-by-step guide to do such a complex task as systematic reviews, especially for scientists, [...]*"), while some still highlight the road ahead ("*[...], but it is still daunting. For instance, the "Extract structured data from*

documents" in "Select" is great in encompassing all the small tasks in code which make it easy to the user/researcher. While "Analysis" and "Synthesis" tasks are not as supported").

5 Discussion and Future Work

Much like Bosman and Kramer's work [14,15], the efforts of this contribution do not end with the publication. The tandem-survey (see footnotes 4, 5) and evaluation remains open for the foreseeable future, encouraging SRA experts from all stages to contribute their expertise for future data collection and integration into the SWARM-SLR framework. Detailing and aligning the different abstraction levels of tasks, steps, results and requirements along the SWARM-SLR workflow requires further interdisciplinary collaboration and evaluation, particularly *Task I: Planning a review*. With additional experts evaluating requirements and providing tool assessments as well as feedback on the workflow, we expect the mapping of accessible and interoperable tools to be achievable and beneficial to scholarly communication and knowledge distribution. Similarly, expanding the evaluation to laymen and new user groups could both aid in differentiating requirement abstraction levels, and reveal additional requirements outside the current consideration. While this work sheds light on the functional requirements these tools and workflows should adhere to, non-functional requirements also affect the success rate of widespread tool and workflow adoption. Our work provides a technical foundation for this, highlighting the opportunity and need for further evaluation, contributions, and collaboration to build a scalable, sustainable, and widely adapted SWARM-SLR.

The current state provides a ready-to-use system for all three major contributions: The 65 *SWARM-SLR Requirements* are a starting point to which none of our survey participants had any further requirements to add, but foreseeable many SLR experts could further strengthen this set. Overall 12 *SWARM-SLR Tools* have been evaluated, only half of them by multiple respondents, and evidently, hundreds more are available for systematic evaluation, a process that could also be further improved by implementing measurable and quantifiable criteria beyond Likert scale agreement. It has been shown that the *SWARM-SLR Workflow* already fulfills most of the requirements by connecting the tools to guide and support the users. Once the tool catalog and requirement set grow further, this workflow similarly needs to be further adapted. The current workflow system, while capable, already has some room for improvement, as evidenced by the continuous improvement during the ongoing Aerospace Knowledge Engineering SWARM-SLR use-case. If this endeavor is continued, we see the potential for eventually (semi-)

automating the SLR workflow. Given a common requirement set like the one presented in this work is established, against which SRA tools are systematically evaluated, the SWARM-SLR tool catalog should grow to provide solutions to most tasks, as already evident. If these tools are made interoperable

in a SWARM-SLR workflow as demonstrated, while further tool development is aimed at evidently less supported tasks, this method should produce a collaboratively developed and improved workflow. This common workflow, potentially specialized to fit certain domains, but unified by the utilization and production of FAIR data artifacts, should lead to domain-independently raising researchers' capabilities to review and manage the dramatically growing scholarly knowledge corpus.

6 Conclusion

In this article, we proposed the Streamlined Workflow Automation for Machine-actionable Systematic Literature Reviews (SWARM-SLR). This includes the requirements, tools measured against these requirements, and a workflow synthesizing these. The requirements were elicited from different domains, streamlining their perspectives and guidelines into a set of 65 requirements. These requirements were evaluated through a tandem-survey alongside potential tools and eventually the implementation of the SWARM-SLR workflow, utilizing a selection of these tools along the research life-cycle from "Formulate research interest" to "Write the literature review". Evidently, these requirements already provide a comprehensive framework to situate a tool's strengths and weaknesses over the SWARM-SLR workflow. We advise to start assessing tools in their usage and development with the SWARM-SLR framework, as well as utilizing these tools and workflow for an efficient AI-supported literature review process.

This initial version of the SWARM-SLR workflow already displays the potential to improve literature-bound knowledge synthesis. The tools evaluated also display specific gaps that can be filled by crowdsourced tool mapping and development. Evidently, all phases, particularly search and retrieval, information extraction, knowledge synthesis, and distribution, can already greatly profit from these advancements. We advise furthering these efforts to develop interfaces between these tools and eventually shape an accessible workflow completely covering all stages. As a result, we envision SWARM-SLR to significantly contribute to digitizing scholarly communication, empower scientists to master the publication flood, to significantly increase AI assistance, and to overall make surveying research more FAIR, collaborative, and effective.

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